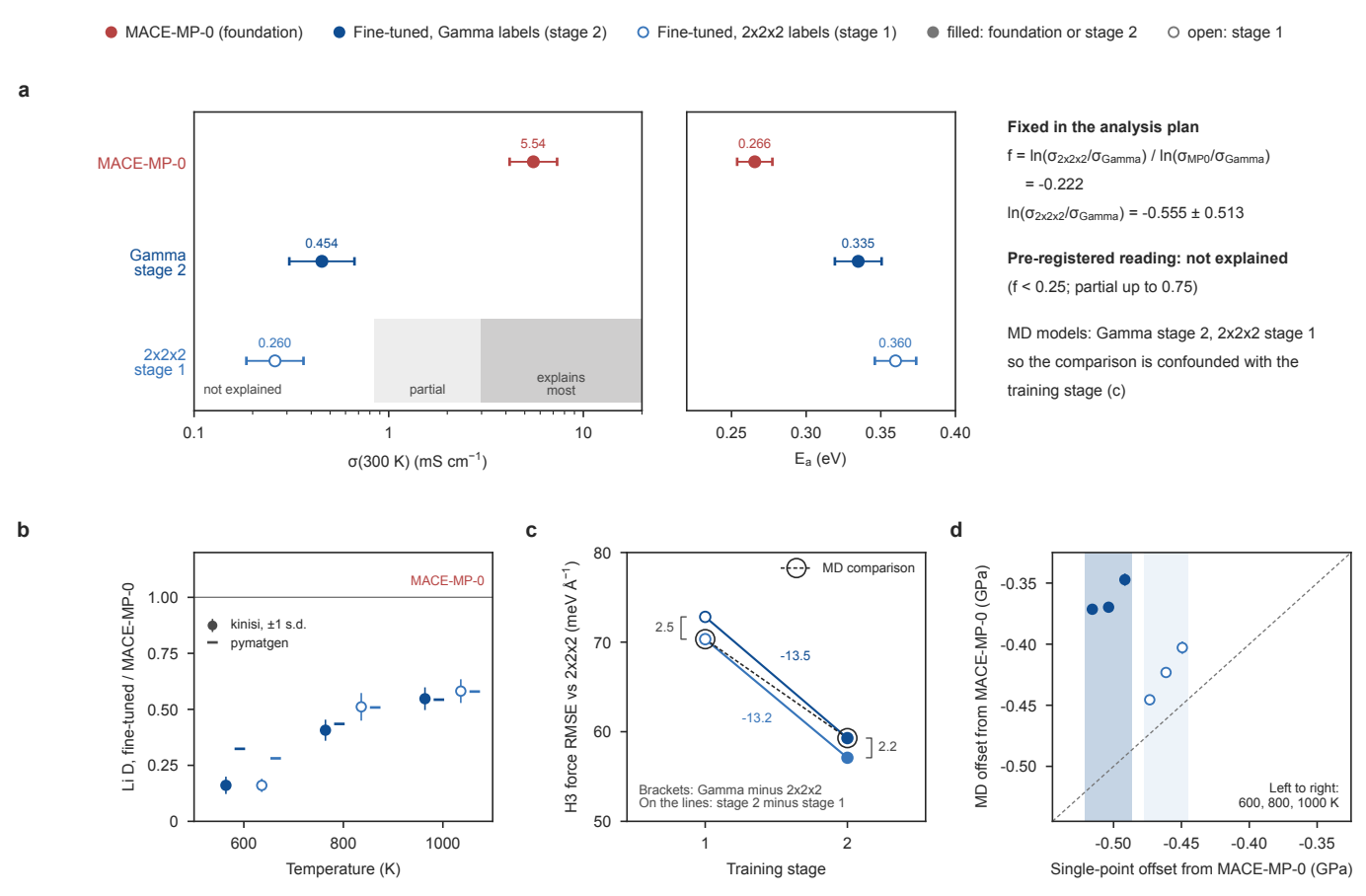

MACE-MP-0 (foundation)
Fine-tuned, Gamma labels (stage 2)
Fine-tuned, 2x2x2 labels (stage 1)
filled: foundation or stage 2
open: stage 1
a
MACE-MP-0
Gamma stage 2
2x2x2 stage 1
5.54
0.454
0.260
not explained
partial
explains most
0.1
1
10
σ(300 K) (mS cm−1)
0.266
0.335
0.360
0.25
0.30
0.35
0.40
Ea (eV)
Fixed in the analysis plan
f = ln(σ2x2x2/σGamma) / ln(σMP0/σGamma)
= -0.222
ln(σ2x2x2/σGamma) = -0.555 ± 0.513
Pre-registered reading: not explained
(f < 0.25; partial up to 0.75)
MD models: Gamma stage 2, 2x2x2 stage 1
so the comparison is confounded with the
training stage (c)
b
MACE-MP-0
kinisi, ±1 s.d.
pymatgen
Li D, fine-tuned / MACE-MP-0
1.00
0.75
0.50
0.25
0
600
800
1000
Temperature (K)
c
MD comparison
2.5
-13.5
-13.2
2.2
Brackets: Gamma minus 2x2x2
On the lines: stage 2 minus stage 1
H3 force RMSE vs 2x2x2 (meV Å−1)
80
70
60
50
1
2
Training stage
d
MD offset from MACE-MP-0 (GPa)
-0.35
-0.40
-0.45
-0.50
Left to right:
600, 800, 1000 K
-0.50
-0.45
-0.40
-0.35
Single-point offset from MACE-MP-0 (GPa)